ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 10	Trace the sequence step by step.
2	**(d)** 14	Pairs: $(1344, 872) \to (872, 472) \to (472, 400) \to (400, 72) \to (328, 72) \to (256, 72) \to (184, 72) \to (112, 72) \to (72, 40) \to (40, 32) \to (32, 8) \to (24, 8) \to (16, 8) \to (8, 8) \to (8, 0)$.
3	**(a)** 8	Pairs: $(912, 720) \to (720, 192) \to (528, 192) \to (336, 192) \to (192, 144) \to (144, 48) \to (96, 48) \to (48, 48) \to (48, 0)$.
4	**(a)** 2075	The algorithm reverses the digits.
5	**(c)** 12218	LCM $= \dfrac{82 \times 298}{2}$.
6	**(d)** 48	$\lfloor \sqrt{2381} \rfloor = 48$.
7	**(d)** 21	Multiples of 3: 15; of 5: 9; of both: 3.
8	**(d)** 32	Pairs: $(2624, 1824) \to (1824, 800) \to (1024, 800) \to (800, 224) \to (576, 224) \to (352, 224) \to (224, 128) \to (128, 96) \to (96, 32) \to (64, 32) \to (32, 32) \to (32, 0)$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$48 = 6 \times 8$ and $18 = 6 \times 3$, and no larger number divides both; R is the definition of gcd.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$29 = 4 \times 7 + 1$, so A is true; $7 = 0 \times 29 + 7$, so R is true; the two statements are unrelated.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	gcd(380, 520): $(520, 380) \to (380, 140) \to (240, 140) \to (140, 100) \to (100, 40) \to (60, 40) \to (40, 20) \to (20, 20) \to (20, 0)$ giving 20; gcd(20, 152): $(152, 20) \to (132, 20) \to (112, 20) \to (92, 20) \to (72, 20) \to (52, 20) \to (32, 20) \to (20, 12) \to (12, 8) \to (8, 4) \to (4, 4) \to (4, 0)$ (1); gcd of the three numbers $= 4$ (1).	1+1
12	Tile side $= \gcd(402, 318)$; $(402, 318) \to (318, 84) \to (234, 84) \to (150, 84) \to (84, 66) \to (66, 18) \to (48, 18) \to (30, 18) \to (18, 12) \to (12, 6) \to (6, 6) \to (6, 0)$ (1); Side 6 cm; Number of tiles $= \dfrac{402 \times 318}{6^2} = 3551$ (1).	1+1
13	$(344, 100) \to (244, 100) \to (144, 100) \to (100, 44) \to (56, 44) \to (44, 12) \to (32, 12) \to (20, 12) \to (12, 8) \to (8, 4) \to (4, 4) \to (4, 0)$; gcd $= 4$ (1); LCM $= \dfrac{344 \times 100}{4} = 8600$ (1).	1+1
14	gcd(280, 480): $(480, 280) \to (280, 200) \to (200, 80) \to (120, 80) \to (80, 40) \to (40, 40) \to (40, 0)$ giving 40; gcd(40, 150): $(150, 40) \to (110, 40) \to (70, 40) \to (40, 30) \to (30, 10) \to (20, 10) \to (10, 10) \to (10, 0)$ (1); gcd of the three numbers $= 10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	For $i = 1$ the inner loop runs 11 times, for $i = 2$ it runs 10 times, and so on (1); Count $= 11 + 10 + \cdots + 1 = \frac{11 \times 12}{2}$ (1); Count $= 66$ (1).	1+1+1
16	Squares added: 9, 36, 81 (1); Total $= 9 + 36 + 81$ (1); Final total $= 126$ (1).	1+1+1
17	Sequence: 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 15$ (2).	1+2
	OR: For $i = 1$ the inner loop runs 6 times, for $i = 2$ it runs 5 times, and so on (1); Count $= 6 + 5 + \cdots + 1 = \frac{6 \times 7}{2}$ (1); Count $= 21$ (1).	1+1+1
18	Squares added: 16, 64 (1); Total $= 16 + 64$ (1); Final total $= 80$ (1).	1+1+1
	OR: Sequence: 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 7$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) $6 + 7 = 13$: write 3, carry 1 (1). (ii) $8 + 4 + 1 = 13$: write 3, carry 1 (1). (iii) $5 + 9 + 1 = 15$: write 5, carry 1 (1). (iv) The last carry becomes the thousands digit, so the sum is 1533 (1) and 3 carries are produced (1).	1+1+1+2
20	$(340, 250) \to (250, 90) \to (160, 90) \to (90, 70) \to (70, 20) \to (50, 20) \to (30, 20) \to (20, 10) \to (10, 10) \to (10, 0)$ (1); Largest common divisor $= \gcd = 10$ (1); Every common divisor divides 10 (1); Common divisors: 1, 2, 5, 10 (2).	1+1+1+2
	OR: gcd(864, 320): $(864, 320) \to (544, 320) \to (320, 224) \to (224, 96) \to (128, 96) \to (96, 32) \to (64, 32) \to (32, 32) \to (32, 0)$ giving 32 (1); gcd(32, 212): $(212, 32) \to (180, 32) \to (148, 32) \to (116, 32) \to (84, 32) \to (52, 32) \to (32, 20) \to (20, 12) \to (12, 8) \to (8, 4) \to (4, 4) \to (4, 0)$ (2); gcd of the three numbers $= 4$ (2).	1+2+2

Mistakes to avoid:

- Forgetting to add the carry from the previous column in column addition.
- Counting a carry when the column sum is exactly 9 (a carry needs a sum of 10 or more).
- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).
- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3	Pairs: $(615, 410) \to (410, 205) \to (205, 205) \to (205, 0)$.
2	**(c)** 6372	The algorithm reverses the digits.
3	**(c)** 960	LCM $= \dfrac{60 \times 192}{12}$.
4	**(d)** 46	$\lfloor \sqrt{2161} \rfloor = 46$.
5	**(d)** 26	Multiples of 3: 18; of 5: 11; of both: 3.
6	**(d)** 17	Pairs: $(3502, 1071) \to (2431, 1071) \to (1360, 1071) \to (1071, 289) \to (782, 289) \to (493, 289) \to (289, 204) \to (204, 85) \to (119, 85) \to (85, 34) \to (51, 34) \to (34, 17) \to (17, 17) \to (17, 0)$.
7	**(a)** 9	Trace the sequence step by step.
8	**(b)** 13	Pairs: $(1790, 1060) \to (1060, 730) \to (730, 330) \to (400, 330) \to (330, 70) \to (260, 70) \to (190, 70) \to (120, 70) \to (70, 50) \to (50, 20) \to (30, 20) \to (20, 10) \to (10, 10) \to (10, 0)$.
9	**(c)** A is true but R is false.	$36 = 12 \times 3,\ 60 = 12 \times 5$, so A is true; R is false because the gcd never exceeds the smaller number.
10	**(d)** A is false but R is true.	There are 6 divisors, so A is false; R lists them correctly.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tile side $= \gcd(380, 171)$; $(380, 171) \to (209, 171) \to (171, 38) \to (133, 38) \to (95, 38) \to (57, 38) \to (38, 19) \to (19, 19) \to (19, 0)$ (1); Side 19 cm; Number of tiles $= \dfrac{380 \times 171}{19^2} = 180$ (1).	1+1
12	$(336, 138) \to (198, 138) \to (138, 60) \to (78, 60) \to (60, 18) \to (42, 18) \to (24, 18) \to (18, 6) \to (12, 6) \to (6, 6) \to (6, 0)$; gcd $= 6$ (1); LCM $= \dfrac{336 \times 138}{6} = 7728$ (1).	1+1
13	gcd(260, 560): $(560, 260) \to (300, 260) \to (260, 40) \to (220, 40) \to (180, 40) \to (140, 40) \to (100, 40) \to (60, 40) \to (40, 20) \to (20, 20) \to (20, 0)$ giving 20; gcd(20, 170): $(170, 20) \to (150, 20) \to (130, 20) \to (110, 20) \to (90, 20) \to (70, 20) \to (50, 20) \to (30, 20) \to (20, 10) \to (10, 10) \to (10, 0)$ (1); gcd of the three numbers $= 10$ (1).	1+1
14	Tile side $= \gcd(215, 301)$; $(301, 215) \to (215, 86) \to (129, 86) \to (86, 43) \to (43, 43) \to (43, 0)$ (1); Side 43 cm; Number of tiles $= \dfrac{215 \times 301}{43^2} = 35$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	For $i = 1$ the inner loop runs 5 times, for $i = 2$ it runs 4 times, and so on (1); Count $= 5 + 4 + \cdots + 1 = \frac{5 \times 6}{2}$ (1); Count $= 15$ (1).	1+1+1
16	Squares added: 9, 36, 81 (1); Total $= 9 + 36 + 81$ (1); Final total $= 126$ (1).	1+1+1
17	Sequence: 21, 64, 32, 16, 8, 4, 2, 1 (1); Number of steps $= 7$ (2).	1+2
	OR: For $i = 1$ the inner loop runs 12 times, for $i = 2$ it runs 11 times, and so on (1); Count $= 12 + 11 + \cdots + 1 = \frac{12 \times 13}{2}$ (1); Count $= 78$ (1).	1+1+1
18	Squares added: 9, 36 (1); Total $= 9 + 36$ (1); Final total $= 45$ (1).	1+1+1
	OR: Sequence: 19, 58, 29, 88, 44, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 20$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $(84, 36) \to (48, 36)$ (1). (ii) $(48, 36) \to (12, 36)$ (1). (iii) $(12, 36) \to (12, 24) \to (12, 12)$, so the gcd is 12 (1). (iv) Side $= 12$ cm (1); number of tiles $= \dfrac{84}{12} \times \dfrac{36}{12} = 7 \times 3 = 21$ (1).	1+1+1+2

20	$(693, 528) \to (528, 165) \to (363, 165) \to (198, 165) \to (165, 33) \to (132, 33) \to (99, 33) \to (66, 33) \to (33, 33) \to (33, 0)$ (1); Largest common divisor $= \gcd = 33$ (1); Every common divisor divides 33 (1); Common divisors: 1, 3, 11, 33 (2).	1+1+1+2
	OR: gcd(540, 870): $(870, 540) \to (540, 330) \to (330, 210) \to (210, 120) \to (120, 90) \to (90, 30) \to (60, 30) \to (30, 30) \to (30, 0)$ giving 30 (1); gcd(30, 159): $(159, 30) \to (129, 30) \to (99, 30) \to (69, 30) \to (39, 30) \to (30, 9) \to (21, 9) \to (12, 9) \to (9, 3) \to (6, 3) \to (3, 3) \to (3, 0)$ (2); gcd of the three numbers $= 3$ (2).	1+2+2

Mistakes to avoid:

- Forgetting to add the carry from the previous column in column addition.
- Counting a carry when the column sum is exactly 9 (a carry needs a sum of 10 or more).
- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).
- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 4452	LCM $= \dfrac{212 \times 84}{4}$.
2	**(a)** 37	$\lfloor \sqrt{1427} \rfloor = 37$.
3	**(a)** 28	Multiples of 3: 20; of 5: 12; of both: 4.
4	**(d)** 50	Pairs: $(2500, 1450) \to (1450, 1050) \to (1050, 400) \to (650, 400) \to (400, 250) \to (250, 150) \to (150, 100) \to (100, 50) \to (50, 50) \to (50, 0)$.
5	**(b)** 21	Trace the sequence step by step.
6	**(b)** 14	Pairs: $(1962, 1044) \to (1044, 918) \to (918, 126) \to (792, 126) \to (666, 126) \to (540, 126) \to (414, 126) \to (288, 126) \to (162, 126) \to (126, 36) \to (90, 36) \to (54, 36) \to (36, 18) \to (18, 18) \to (18, 0)$.
7	**(d)** 9	Pairs: $(1353, 561) \to (792, 561) \to (561, 231) \to (330, 231) \to (231, 99) \to (132, 99) \to (99, 33) \to (66, 33) \to (33, 33) \to (33, 0)$.
8	**(c)** 9578	The algorithm reverses the digits.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Digit 7 is written and 1 is carried to the tens column because the column sum exceeds 9.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R is about a different number and does not explain A (the reason for A is that 6 divides 24).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(287, 245) \to (245, 42) \to (203, 42) \to (161, 42) \to (119, 42) \to (77, 42) \to (42, 35) \to (35, 7) \to (28, 7) \to (21, 7) \to (14, 7) \to (7, 7) \to (7, 0)$; gcd $= 7$ (1); LCM $= \dfrac{287 \times 245}{7} = 10045$ (1).	1+1
12	gcd(128, 192): $(192, 128) \to (128, 64) \to (64, 64) \to (64, 0)$ giving 64; gcd(64, 144): $(144, 64) \to (80, 64) \to (64, 16) \to (48, 16) \to (32, 16) \to (16, 16) \to (16, 0)$ (1); gcd of the three numbers $= 16$ (1).	1+1
13	Tile side $= \gcd(128, 164)$; $(164, 128) \to (128, 36) \to (92, 36) \to (56, 36) \to (36, 20) \to (20, 16) \to (16, 4) \to (12, 4) \to (8, 4) \to (4, 4) \to (4, 0)$ (1); Side 4 cm; Number of tiles $= \dfrac{128 \times 164}{4^2} = 1312$ (1).	1+1
14	$(399, 308) \to (308, 91) \to (217, 91) \to (126, 91) \to (91, 35) \to (56, 35) \to (35, 21) \to (21, 14) \to (14, 7) \to (7, 7) \to (7, 0)$; gcd $= 7$ (1); LCM $= \dfrac{399 \times 308}{7} = 17556$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	For $i = 1$ the inner loop runs 8 times, for $i = 2$ it runs 7 times, and so on (1); Count $= 8 + 7 + \cdots + 1 = \frac{8 \times 9}{2}$ (1); Count $= 36$ (1).	1+1+1
16	Squares added: 16 (1); Total $= 16$ (1); Final total $= 16$ (1).	1+1+1
17	Sequence: 30, 15, 46, 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 18$ (2).	1+2
	OR: For $i = 1$ the inner loop runs 9 times, for $i = 2$ it runs 8 times, and so on (1); Count $= 9 + 8 + \cdots + 1 = \frac{9 \times 10}{2}$ (1); Count $= 45$ (1).	1+1+1
18	Squares added: 9, 36, 81 (1); Total $= 9 + 36 + 81$ (1); Final total $= 126$ (1).	1+1+1
	OR: Sequence: 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 12$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $29 = 4 \times 7 + 1$, so 1 (1). (ii) $100 = 14 \times 7 + 2$, so 2 (1). (iii) Monday is 1, and $(1 + 2) \bmod 7 = 3$, which is Wednesday (1). (iv) $45 \bmod 7 = 3$ (1); $(4 + 3) \bmod 7 = 0$, which is Sunday (1).	1+1+1+2

20	$(553, 385) \to (385, 168) \to (217, 168) \to (168, 49) \to (119, 49) \to (70, 49) \to (49, 21) \to (28, 21) \to (21, 7) \to (14, 7) \to (7, 7) \to (7, 0)$ (1); Largest common divisor $= \gcd = 7$ (1); Every common divisor divides 7 (1); Common divisors: 1, 7 (2).	1+1+1+2
	OR: gcd(430, 258): $(430, 258) \to (258, 172) \to (172, 86) \to (86, 86) \to (86, 0)$ giving 86 (1); gcd(86, 222): $(222, 86) \to (136, 86) \to (86, 50) \to (50, 36) \to (36, 14) \to (22, 14) \to (14, 8) \to (8, 6) \to (6, 2) \to (4, 2) \to (2, 2) \to (2, 0)$ (2); gcd of the three numbers $= 2$ (2).	1+2+2

Mistakes to avoid:

- Forgetting to add the carry from the previous column in column addition.
- Counting a carry when the column sum is exactly 9 (a carry needs a sum of 10 or more).
- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).
- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND